

CredLock

Collateral safety for Creditcoin financing.

An asset gets pledged as collateral on one chain, then shows up at Creditcoin asking for financing.
The second lender cannot see the first pledge.

CredLock checks before Creditcoin lends. Same pattern for RWA collateral: never finance what is already pledged elsewhere.

Built for lenders and Creditcoin credit apps that need a gate before money moves.

Pledged -> BLOCKED

Pledged on Sepolia.
Attestcoin proves it.
Creditcoin verifies inside Borrow.
Reverts: AssetEncumbered.

Clear -> ALLOWED

Clear on Sepolia.
Attestcoin proves it.
Creditcoin verifies inside Borrow.
Financing executes.

One rule: the financing transaction carries its own proof, or it does not happen.

HOW IT WORKS

Borrow carries the proof.

Not trusted from the frontend alone. The Borrow transaction brings a fresh Attestcoin proof to the gate. The gate verifies it inline and decides from the proven fact.

1 Sepolia registry

SourcePledgeRegistry emits AssetRegistered (clear) and AssetPledged (encumbered) for one asset id.

2 Attestcoin proof

Merkle inclusion plus continuity proof for the newest fact, built with the usc-sdk proof builder.

3 Creditcoin gate

requestFinancingWithProof verifies through the BlockProver precompile (0xFD2) and checks the trusted emitter, chain, and event.

4 Decision

Clear fact: records ALLOW, executes financing. Pledged fact: reverts AssetEncumbered. A recorded BLOCK can never reopen.

DEPLOYED NOW

Gate - Creditcoin 102031

0xF1028099d9CeE27b355159beCD111c8f9A409F67

Registry - Sepolia

0xae543bb778df66774585b99d9135676bc5d99c2a

Source chainKey 1. 20 Foundry tests, all green.

Solidity + Foundry. EvmV1Decoder event checks. Next.js + wagmi UI, every write signed by the user wallet. No server key, no mocks on the verification path.

PROOF IT WORKS

Two assets, two outcomes.

SCENARIO A - CLEAR ASSET BORROWS

Asset 0x6349...0ae9e3. Registered on Sepolia, never pledged. Borrow carried the fresh registration proof. Verified on-chain, financing executed, ALLOW recorded.

Register (Sepolia)

0x1f3fee534c5f757ebd70b979102d43ef01d6baea07a921ff59db72b3c6c1c6d4

Borrow (Creditcoin, block 5470652)

0xab138a65bc019554e1bf5a161451ece5b977791df90c0e2610e8f7188fd16c7d

SCENARIO B - PLEDGED ASSET REVERTS

Asset 0x8bfc...024d9 (uyscutty). Registered, then pledged on Sepolia. Borrow carried the fresh pledge proof. The gate reverted with AssetEncumbered, proven on a live node call and covered by the revert tests. No path through this gate can finance it.

Pledge (Sepolia, block 11684010)

0xd0625d429caab95496629bde3c50c865ac9d4c04bc2e2881d7e39fcd2f23c107

Revert reason

AssetEncumbered (0x01d7f74f) - live call + 20/20 forge tests

LINKS

Live app

<https://credlock-neon.vercel.app/>

Repository

<https://github.com/uyscutty/CredLock>

Gate contract

[https://creditcoin-testnet.blockscout.com/
address/0xF1028099d9CeE27b355159beCD111c8f9A409F67](https://creditcoin-testnet.blockscout.com/address/0xF1028099d9CeE27b355159beCD111c8f9A409F67)

Registry contract

[https://sepolia.etherscan.io/
address/0xae543bb778df66774585b99d9135676bc5d99c2a](https://sepolia.etherscan.io/address/0xae543bb778df66774585b99d9135676bc5d99c2a)

Borrow tx (A)

[https://creditcoin-testnet.blockscout.com/
tx/0xab138a65bc019554e1bf5a161451ece5b977791df90c0e2610e8f7188fd16c7d](https://creditcoin-testnet.blockscout.com/tx/0xab138a65bc019554e1bf5a161451ece5b977791df90c0e2610e8f7188fd16c7d)

Register tx (A)

[https://sepolia.etherscan.io/
tx/0x1f3fee534c5f757ebd70b979102d43ef01d6baea07a921ff59db72b3c6c1c6d4](https://sepolia.etherscan.io/tx/0x1f3fee534c5f757ebd70b979102d43ef01d6baea07a921ff59db72b3c6c1c6d4)

Pledge tx (B)

[https://sepolia.etherscan.io/
tx/0xd0625d429caab95496629bde3c50c865ac9d4c04bc2e2881d7e39fcd2f23c107](https://sepolia.etherscan.io/tx/0xd0625d429caab95496629bde3c50c865ac9d4c04bc2e2881d7e39fcd2f23c107)

Proof builder

<https://prover.cc3-testnet.creditcoin.network>